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13 August 1965
SUPERSEDING
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MILITARY SPECIFICATION

BRASS, POWDER (FOR SINTERINGS)

1. SCOPE

1.1 Scope. This specification covers requirements for brass powder to be used in sinterings.

1.2 Classification.

1.2.1 Compositions. The material covered by this specification shall be furnished in the composition hereinafter specified (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151 - Metals; Test Methods

MILITARY

MIL-STD-129-Marking for Shipment and Storage

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FSC MISC

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

B212-48 - Apparent Density of Metal Powders
B213-48 - Flow Rate of Metal Powders
B312-58T - Green Strength of Compacted Metal Powder Specimens

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa., 19103.)

3. REQUIREMENTS

3.1 Chemical requirements. The material shall conform to the chemical requirements shown in table I.

Table I - Chemical composition, percent

Composition	Copper, min.	Copper, max.	Lead, max.	Iron, max.	Zinc	Total other elements, max.
1	88.0	91.0	0.05	0.05	Rem.	0.15
3	77.0	80.0	1.0-2.0	0.05	Rem.	0.15
5	68.5	71.5	0.07	0.05	Rem.	0.15

3.1.1 Analysis shall be made regularly only for the elements specifically mentioned in table I. If, however, the presence of other elements is suspected or indicated in the course of routine analysis, further analysis shall be made to determine that the total of other elements is not in excess of the limits specified.

3.2 Density. The material shall meet the apparent density requirements as indicated in table II.

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Table II - Apparent density of powders

Composition	Apparent density (grams/cc)
1	3.00 - 3.20
3	3.00 - 3.20
5	2.80 - 3.00

3.3 Flow rate. The material in the unlubricated condition shall be free flowing and shall meet the requirements of table III.

Table III - Flow rate

Composition	Flow rate (seconds/50 grams)
1	24 - 26
3	23 - 27
5	27 - 31

3.4 Green strength. The green strength of the compacted powder shall be as specified in the contract or order.

3.5 Workmanship. The material shall be irregular granules with flaky or acicular projections, rough surfaced, and shall be free from foreign materials which would make the powder unsuitable for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified in the contract or order, a lot shall consist of material of the same composition, made by the same manufacturing process, and submitted for inspection at one time.

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4.3 Sampling. A sample of the entire cross-section of the stream of thoroughly blended powder flowing from a blender shall be taken half-way during the filling of the first shipping container. The filling of the containers shall be continued until half the burden of the blender has been discharged, at which time another sample, including the entire cross-section of the discharge stream, shall again be taken. The filling shall be continued until half-way through the filling of the last shipping container at which time a sample of the entire cross-section of the discharge stream shall again be taken. The samples shall not be mixed. In cases where the lot consists of less than three containers, each container shall be sampled by taking a portion of the full cross-section of the blender discharge stream. Each sample shall be reduced by passing it through a sample splitter until the proper size samples for chemical, density determination, flow rate test and green strength test are obtained.

4.3.1 For chemical analysis. From each sample taken in accordance with 4.3, a specimen of 5 grams shall be obtained for government analysis. The specimens from each lot shall not be mixed.

4.3.2 For density determination. One density determination shall be performed for each sample obtained in accordance with 4.3.

4.3.3 For flow rate test. One flow rate test shall be performed for each sample obtained in accordance with 4.3.

4.3.4 For green strength test. One green strength test shall be performed for each sample obtained in accordance with 4.3.

4.4 Test procedures.

4.4.1 Chemical analysis. The samples selected in accordance with 4.3.1 shall be analyzed by the wet chemical method in accordance with method 111 of Fed. Test Method Std. No. 151 or the spectrochemical method in accordance with method 112 of Fed. Test Method Std. No. 151 to determine conformance with 3.1. In case of dispute, the analysis by the wet method (method 111) shall be the basis for acceptance.

4.4.2 Density. The apparent density of the powders shall be determined in accordance with Standard Method of Test for Apparent Density of Metal Powders, ASTM B212-48.

4.4.3 Flow rate. The flow rate of the powders shall be determined in accordance with Standard Method of Test for Flow Rate of Metal Powders, ASTM B213-48.

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4.4.4 Green strength. The green strength of the compacted powders shall be determined in accordance with the Tentative Method of Test for Green Strength of Compacted Metal Powder Specimens, ASTM B312-58T.

4.5 Examination.

4.5.1 The samples taken in accordance with 4.3 shall be examined for compliance with the requirements for workmanship (see 3.5). In addition, all other material in the lot shall be subject to examination for compliance with the requirements for workmanship.

4.5.2 Preparation for shipment. Examination of the packaging and packing, and marking for shipment shall be made for conformance with the requirements of section 5.

4.6 Test failures. A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.4, subject to the provisions of the section on "Rejection and Retests" of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

5.1 Packaging - levels A, B and C. The powder shall be packaged in 110 pound net metallic pails or 660 pounds net metallic drums with crimp seal or soldered lid.

5.2 Packing - levels A, B and C The product shall be separated by composition and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, in such a manner as to insure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection from normal hazards of transportation.

5.3 Marking. In addition to any special marking required in the contract or order (see 6.2), marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The brass powders covered by this specification are intended to be used in normal powder metallurgy fabrication involving compacting and sintering for parts in time fuzes, structural parts of fire control instruments, and other miscellaneous powder metallurgy applications.

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6.1.1 Composition 1 is intended mainly for use as an alternate material for gilding metal rotating bands.

6.1.2 Composition 3 is a leaded 80-20 brass. It is intended for use when ease of machining is desired.

6.1.3 Composition 5 is intended for use when the sintering will be subjected to cold working or deep drawing. Compacts made from powder of this composition can have mechanical properties comparable to the cast-wrought material of the same composition in the annealed condition. To attain such property levels the "nitrogen sintering method" must be used. This composition is recommended for highly stressed structural applications.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Composition required (see 1.2.1).
(d) Green strength requirements (see 3.4).
(d) Whether material is to be packaged and packed for level A, B or C (see 5.1 and 5.2).
(e) Special marking, if required.

6.3 The requirements for packaging (see 5.1) and for packing (see 5.2) and marking for shipment (see 5.3) specified herein apply to direct shipment for Government activities and apply also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

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6.4 Design information. The processing steps - briquetting and sintering - are closely controlled because the ultimate properties of the parts largely depend upon these controls. The briquetting pressure is the most important variable in determining the density of the finished part. In the parts covered by the specifications dealing with structural parts, this density may vary from 72 to over 99 percent of the density of cast or wrought parts of the same composition. Densities lower than those of cast or wrought parts cause the parts to be inherently porous and differentiate them from cast or wrought materials. It has a pronounced effect upon the properties of the parts.

Variables in the sintering step are sintering temperature, time, and atmosphere. For many parts there is an optimum sintering temperature range in which parts of a given composition will reach maximum mechanical properties. Sintering below or above this range may lower the mechanical properties without much effect upon their density or chemical composition. This optimum temperature range may however not be the most practical if the dimensional changes during sintering in this temperature range cannot be satisfactorily controlled. Sintering time has, in general, a lesser effect upon properties, but too short a sintering time may also seriously lower mechanical properties. Finally, the sintering atmosphere may affect the mechanical properties and must therefore be controlled.

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